

1. Consider a network address 10.0.0.0/8. You have to create subnets each of which contains 0 maximum 3500 hosts. Make sure the loss of host address is minimum. Find the following information-

a) Total Number of sub-networks: _____ 4096 (2^{12}) _____

b) Minimum number of wastage of usable hosts per subnet: _____ 4094-3500=594 _____

c) Suppose you are the network administrator of subnet number _326_. (subnet 0 is considered as subnet number 1). Find the following information of your network.

i) Network ID: _____ 10.19.96.0 _____

ii) Subnet Mask: _____ 255.255.240.0 _____

iii) Broadcast Address: _____ 10.19.111.255 _____

iv) First usable Host Address: _____ 10.19.96.1 _____

v) Last usable Host Address: _____ 10.19.111.254 _____

2. Your PC IP address is 172.16.93.4 and subnet mask is 255.255.248.0. Find the following information-

a) Which subnet your PC belongs to: _____ 172.16.88.0/21 _____

b) What will be the 258th usable host address of the next subnet? _____ 172.16.97.4/21 _____

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Department of Computer Science and Engineering
Cisco Networking Academy Program (CNAP)
Skill Final Examination, 2009

ID:

Name:

Batch:

Set:

1. Consider a network address 172.16.190.57/18. You have to create subnets each of which contains maximum 115 hosts. Make sure the loss of host address is minimum. Find the following information-
 - a) Total Number of sub-networks:
_____ $2^7=128$ _____
 - b) Minimum number of wastage of usable hosts per subnet: _____ $2^7=128-2-115=11$ _____
 - c) Suppose you are the network administrator of subnet _____ 109 _____ (subnet 0 is considered as subnet number 1). Find the following information of your network.
 - i) Network ID: _____ 172.16.182.128 _____
 - ii) Subnet Mask: _____ 255.255.255.128 _____
 - iii) Broadcast Address: _____ 172.16.182.255 _____
 - iv) First usable Host Address: _____ 172.16.182.129 _____
 - v) Last usable Host Address: _____ 172.16.182.254 _____
2. Your PC IP address is 192.168.1.254 and subnet mask is 255.255.255.240. Find the following information-
 - a) Which subnet your PC belongs to: _____ 192.168.1.240 _____
 - b) Which is the last address of the subnet to which your PC belongs: _____ 192.168.1.254 _____
 - c) If it were made with Class C addressing, how many valid host addresses would waste:
_____ $254-224=30$ _____

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ID:

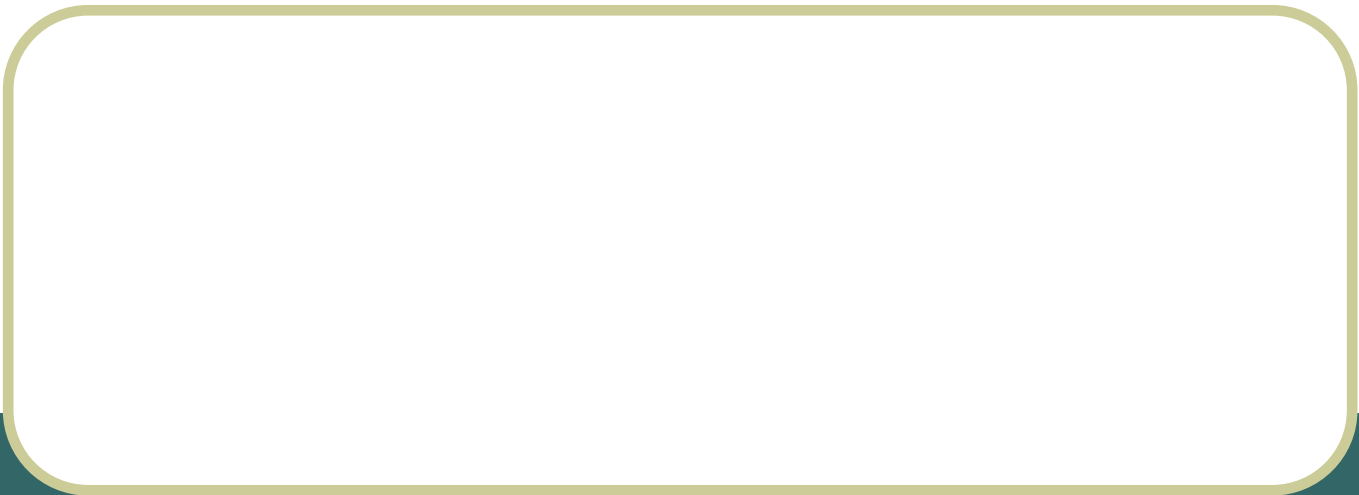
Name:

Batch:

Set:

1. Consider a network address 172.16.0.0/16. You have to create subnets each of which contains at least 22 hosts. Make sure the loss of host address is minimum. Find the following information-
 - a) Total Number of sub-networks: _____
 - b) Total number of wastage of usable hosts per subnet: _____
 - c) Suppose you are the network administrator of subnet _____ (subnet 0 is considered as subnet number 1). Find the following information of your network.
 - i) Network ID: _____
 - ii) Subnet Mask: _____
 - iii) Broadcast Address: _____
 - iv) First usable Host Address: _____
 - v) Last usable Host Address: _____
2. Your PC IP address is 172.16.155.10 and subnet mask is 255.255.224.0. Find the following information-
 - a) Which subnet your PC belongs to: _____
 - b) Which is the last address of the subnet to which your PC belongs: _____
 - c) If it were made with Class B addressing, how many valid host addresses would waste:

IP4 Subnetting



The Problem

- Let an ISP is assigned an address block, and consider one of the address from that block is 20.69.0.0 /10
 - Subnet the network so that each subnet has at least 3600 hosts (Host IP loss should be as minimum as possible)
 1. How many subnets?
 2. Consider subnet # **373**, and Find
 - a. Network Address of the subnet
 - b. First, Last host address
 - c. Broadcast address
 - d. Unnecessary Host address in the subnet
 3. How many Host address losses for subnetting?

Solution

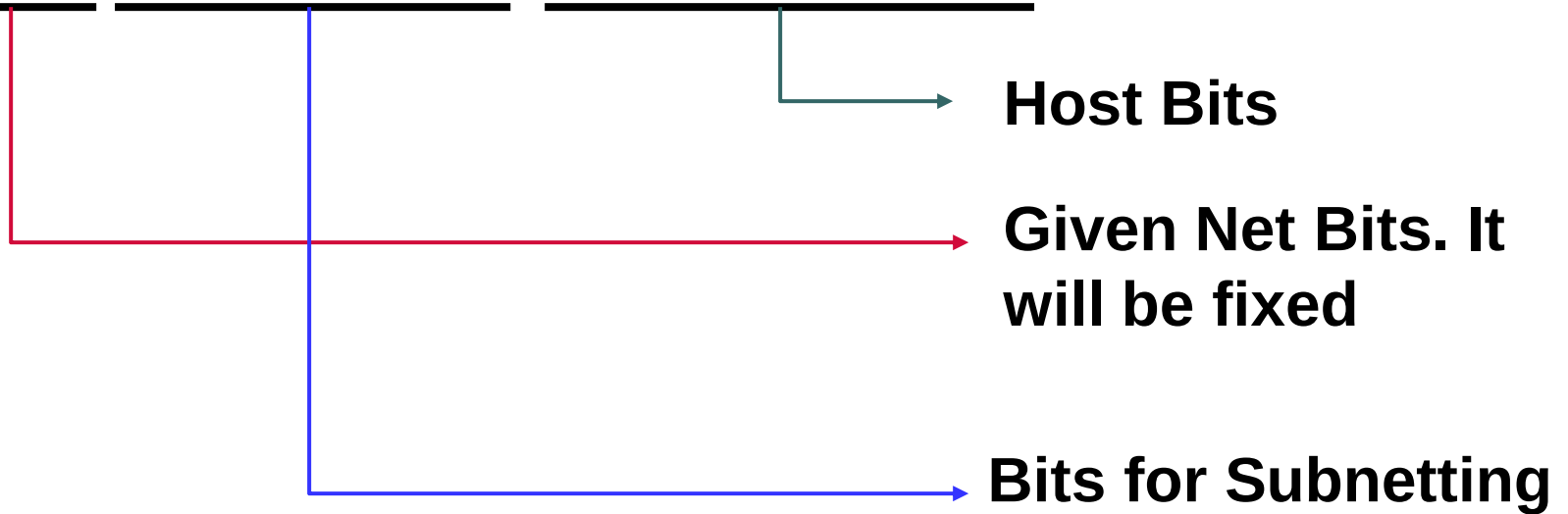
- Given Address: 20.69.0.0 = **20.01000101.0.0**
- Given Mask : 255.192.0.0 = **255.11000000.0.0**
- Given Network : 20.64.0.0 = **20.01000000.0.0**
- Net Bits in given Address = **10**
- Required 3600 Hosts
 - Required Host Bits = **12** (**why?** $2^{11} = 2048$, $2^{12} = 4096$)
- Bits remain for subnetting = $32 - 10 - 12 = \mathbf{10}$

ANS

- 1. Total subnets = $2^{\text{Subnet Bits}} = 2^{10} = 1024$

Solution

- **20.01**000000.00000000.00000000



- Binary for **373**

- 256 128 64 32 16 8 4 2 1
- 1 0 1 1 1 0 1 0 1 (9 bits only)

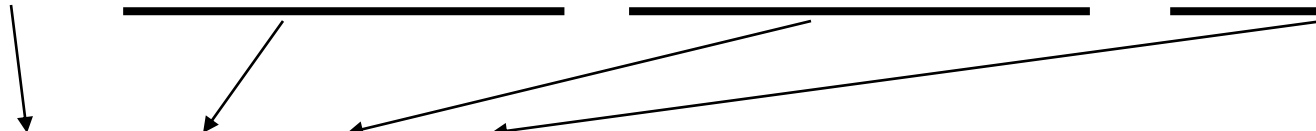
Solution

- Now consider Subnet Bits, and Host Bits just as place holder
- **20** . **01** _ _ _ _ _ . _ _ _ _ _
- 10 places for subnet bits
 - But the binary of **373** has only **9** bits
 - Put **One** additional **0** before the binary number
 - And place it in the place holder of Subnet bits
- **20** . **01** 0 1 0 1 1 1 . 0 1 0 1 _ _ _ _ _
- New Network Bits (Total 20 bits)
- **20** . **01** 0 1 0 1 1 1 . 0 1 0 1 _ _ _ _ _

Solution

- Now for **Subnet** address place **0** on host bits

● **20. 01 0 1 0 1 1 1 . 0 1 0 1 0 0 0 0 . 0 0 0 0 0 0 0 0**

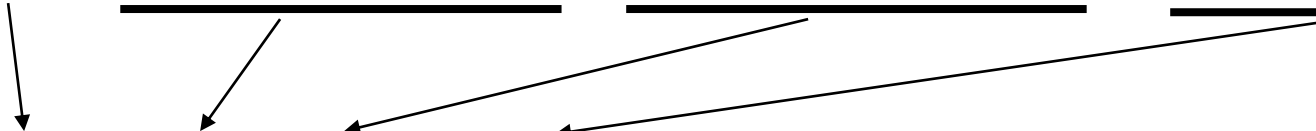


The diagram shows four arrows pointing from the binary address to the decimal address below. The first arrow points from the '20' to the first octet '20'. The second arrow points from the first '0' of the second octet to the '87'. The third arrow points from the '0' of the third octet to the '80'. The fourth arrow points from the first '0' of the fourth octet to the '0'.

- **20 . 87 . 80 . 0 / 20** is the **Subnet Address**

- For **Broadcast** address place **1** on host bits

● **20. 01 0 1 0 1 1 1 . 0 1 0 1 1 1 1 1 . 1 1 1 1 1 1 1 1**



The diagram shows four arrows pointing from the binary address to the decimal address below. The first arrow points from the '20' to the first octet '20'. The second arrow points from the first '0' of the second octet to the '87'. The third arrow points from the '1' of the third octet to the '95'. The fourth arrow points from the first '1' of the fourth octet to the '255'.

- **20 . 87 . 95 . 255** is the **Subnet Address**

Solution

- For First Address just **add 1** with **Net** address

$$\begin{array}{r} 20 . 87 . 80 . 0 \\ + 1 \\ \hline 20 . 87 . 80 . 1 \end{array}$$

- For Last Address just **subtract 1** from **broadcast** address

$$\begin{array}{r} 20 . 87 . 95 . 255 \\ - 1 \\ \hline 20 . 87 . 95 . 254 \end{array}$$

Solution

2.d) Host IP in each subnet = $(2^{12} - 2) = 4094$

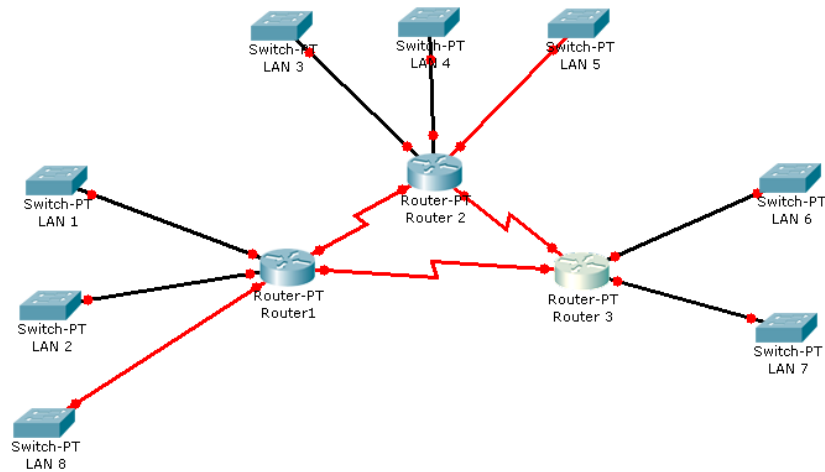
Unnecessary Host IPs = $4094 - 3600 = 494$

3. Host IP losses for subnetting

Total Host IPs in Original Net = $2^{\text{Host Bits}} = 2^{22} - 2 = A$

Host IPs in subnets = Total subnet X Host IP in each subnet
 $= 2^{10} \times (2^{12} - 2) = B$

Losses = $A - B = (2^{22} - 2) - 2^{10} \times (2^{12} - 2)$
 $= 2^{22} - 2^{22} + 2^{11} - 2 = 2046$



Base Address: 172.16.0.0/16

LAN 1: 1000 hosts
LAN 2: 200 hosts
LAN 3: 2000 hosts
LAN 4: 2000 hosts
LAN 5: 500 hosts
LAN 6: 3000 hosts
LAN 7: 100 hosts
LAN 8: 60 hosts

LAN	N/W Address	First Valid Host	Last Valid Host	Broadcast Address
1	172.16.32.0/22	172.16.32.1	172.16.35.254	172.16.35.255
2	172.16.38.0/24	172.16.38.1	172.16.38.254	172.16.38.255
3	172.16.16.0/21	172.16.16.1	172.16.23.254	172.16.23.255
4	172.16.24.0/21	172.16.24.1	172.16.31.254	172.16.31.255
5	172.16.36.0/23	172.16.36.1	172.16.37.254	172.16.37.255
6	172.16.0.0/20	172.16.0.1	172.16.15.254	172.16.15.255
7	172.16.39.0/25	172.16.39.1	172.16.39.126	172.16.39.127
8	172.16.39.128/26	172.16.39.129	172.16.39.190	172.16.39.191

Serial Link	IP Address
Router 1 to Router 2	172.16.39.193/30 and 172.16.39.194/30
Router 2 to Router 3	172.16.39.197/30 and 172.16.39.199/30
Router 1 to Router 3	172.16.39.201/30 and 172.16.39.202/30

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Skill Final Exam: B3 Batch

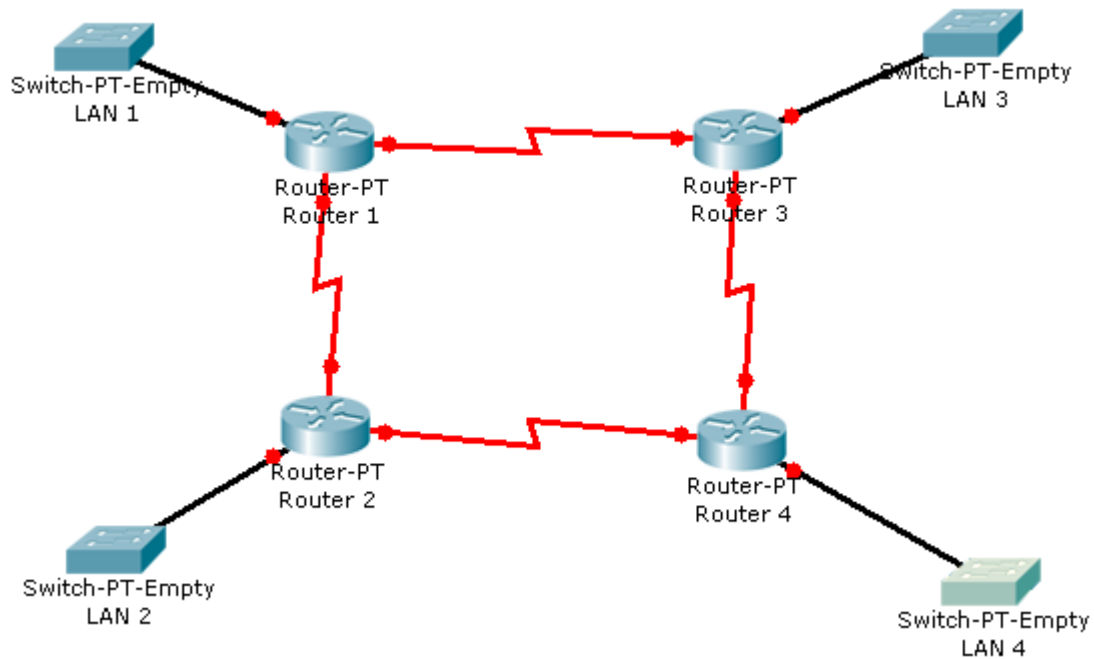
Part 1: VLSM

Time: 25 Minutes

Full Marks: 50

Assign subnets to the following LANs by breaking the base IP Address **180.180.0.0/17** so that they can accommodate the indicated number of hosts with minimum wastage of IP Addresses.

LAN	No. of Hosts	N/W Address	First Valid Host Address	Last Valid Host Address	Broadcast Address
1	1000	180.180.8.0/22	180.180.8.1	180.180.11.254	180.180.11.255
2	1500	180.180.0.0/21	180.180.0.1	180.180.7.254	180.180.7.255
3	30	180.180.14.64/27	180.180.14.65	180.180.14.94	180.180.14.95
4	60	180.180.14.0/26	180.180.14.1	180.180.14.62	180.180.14.63
5	400	180.180.12.0/23	180.180.12.1	180.180.13.254	180.180.13.255
6	2	180.180.14.96/30	180.180.14.97	180.180.14.98	180.180.14.99
7	2	180.180.14.100/30	180.180.14.101	180.180.14.102	180.180.14.103
8	2	180.180.14.104/30	180.180.14.105	180.180.14.106	180.180.14.107



Number of Hosts:

LAN 1: 5	70	1000110	200.200.200.00000000 200.200.200.10000000
LAN 2: 70			
LAN 3: 10			200.200.200.10000000-lan4
LAN 4: 28	28	11100	3 200.200.200.10100000 200.200.200.11000000 200.200.200.11100000

Base Address: 200.200.200.0/24

Procedure:

10	1010	4	200.200.200.10100000-lan 3 200.200.200.10110000
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1. Which LAN has the most number of hosts- start with that LAN. In this case, it's LAN 2.

70 hosts require 7 bits in host portion. So, the prefix will be /25. So, we borrowed 1 bit from the base address- means there will be two subnets.

200.200.200.0/25 → Assign it to LAN 2

200.200.200.128/25

2. Which of the LANs have second most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 4.

28 hosts require 5 bits in host portion. So, the prefix will be /27. So, we borrowed 2 bits from the base address- means there will be four subnets.

200.200.200.128/27 → Assign it to LAN 4

200.200.200.160/27

200.200.200.192/27

200.200.200.224/27

3. Which of the LANs have third most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 3.

10 hosts require 4 bits in host portion. So, the prefix will be /28. So, we borrowed 1 bit from the base address- means there will be two subnets.

200.200.200.160/28 → Assign it to LAN 3

200.200.200.176/28

4. Which of the LANs have fourth most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 1. 5 hosts require 3 bits in host portion. So, the prefix will be /29. So, we borrowed 1 bit from the base address- means there will be two subnets.
200.200.200.176/29 → Assign it to LAN 1
200.200.200.184/29

Now, we have completed subnetting the LANs.

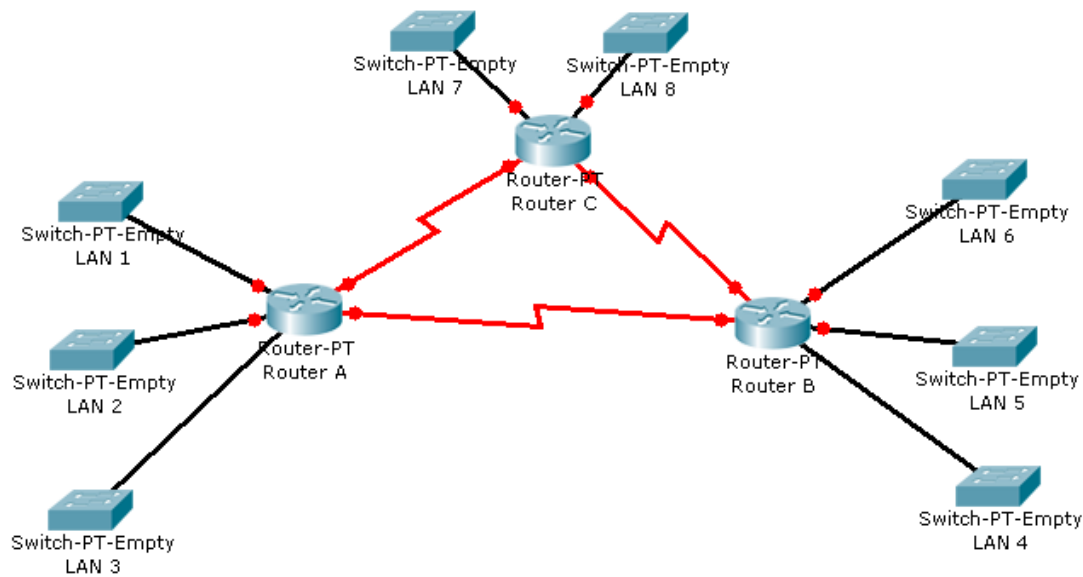
We will now subnet the address for serial links. There are 4 serial links each of them will require 2 hosts. Start with any of them with the base address of the leftover subnet. As we have borrowed 1 bit from the base address, there will be two subnets-

200.200.200.184/30 → Assign it to serial connection 1
200.200.200.188/30 → Assign it to serial connection 2

Now, the subnet 200.200.200.160/27 is fully used. But we still have 2 serial links to have network address. So, we take the next leftover address as the base address- in this case, it is 200.200.200.192/27. For 2 hosts per network, we need 2 bits in host portion. So, 3 bits will be borrowed from the base address.

200.200.200.192/30 → Assign it to serial connection 3
200.200.200.196/30 → Assign it to serial connection 4

200.200.200.200/30
200.200.200.204/30
200.200.200.208/30
200.200.200.212/30
200.200.200.216/30
200.200.200.220/30



Number of Hosts:

LAN 1: 8000
 LAN 2: 2000
 LAN 3: 1000
 LAN 4: 4000
 LAN 5: 4000
 LAN 6: 500
 LAN 7: 254
 LAN 8: 50

Base Address: 180.180.0.0/16

Procedure:

1. Which LAN has the most number of hosts- start with that LAN. In this case, it's LAN 1.

8000 hosts require 13 bits in host portion. So, the prefix will be /19. So, we borrowed 3 bits from the base address- means there will be 8 subnets.

180.180.0.0/19 → Assign it to LAN 1

180.180.32.0/19

180.180.64.0/19

180.180.96.0/19

180.180.128.0/19

180.180.160.0/19

180.180.192.0/19

180.180.224.0/19

2. Which of the LANs have second most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 4, LAN 5. 4000 hosts require 12 bits in host portion. So, the prefix will be /20. So, we borrowed 1 bit from the base address- means there will be 2 subnets.

180.180.32.0/20 → Assign it to LAN 4

180.180.48.0/20 → Assign it to LAN 5

3. Which of the LANs have third most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 2.
2000 hosts require 11 bits in host portion. So, the prefix will be /28. So, we borrowed 1 bit from the base address- means there will be two subnets.

180.180.64.0/21 → Assign it to LAN 2

180.180.72.0/21

4. Which of the LANs have fourth most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 3.
1000 hosts require 10 bits in host portion. So, the prefix will be /22. So, we borrowed 1 bit from the base address- means there will be two subnets.

180.180.72.0/22 → Assign it to LAN 3

180.180.76.0/22

5. Which of the LANs have fourth most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 6.
500 hosts require 9 bits in host portion. So, the prefix will be /23. So, we borrowed 1 bit from the base address- means there will be two subnets.

180.180.76.0/23 → Assign it to LAN 6

180.180.78.0/23

6. Which of the LANs have fourth most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 7.
254 hosts require 8 bits in host portion. So, the prefix will be /24. So, we borrowed 1 bit from the base address- means there will be two subnets.

180.180.78.0/24 → Assign it to LAN 7

180.180.79.0/24

7. Which of the LANs have fourth most number of hosts- start with that LAN having the leftover subnet as the base address. In this case, it's LAN 8.
50 hosts require 6 bits in host portion. So, the prefix will be /26. So, we borrowed 2 bit from the base address- means there will be 4 subnets.

180.180.79.0/26 → Assign it to LAN 8

180.180.79.64/26

180.180.79.128/26

180.180.79.192/26

Now, we have completed subnetting the LANs. We will now subnet the address for serial links. There are 3 serial links each of them will require 2 hosts. Start with any of them with the base address of the leftover subnet. As we have borrowed 1 bit from the base address, there will be two subnets-

180.180.79.64/30 → Assign it to serial connection 1

180.180.79.68/30 → Assign it to serial connection 2

180.180.79.72/30 → Assign it to serial connection 3

180.180.79.76/30

180.180.79.80/30

180.180.79.84/30

180.180.79.88/30

180.180.79.92/30

180.180.79.96/30

180.180.79.100/30

180.180.79.104/30

180.180.79.108/30

180.180.79.112/30

180.180.79.116/30
180.180.79.120/30
180.180.79.124/30

* Given Lan number and number of hosts under each Lan:—

LAN-1	10 hosts
LAN-2	80 hosts
LAN-3	3000 hosts
LAN-4	3000 hosts
LAN-5	20 hosts
LAN-6	1500 hosts
LAN-7	2 hosts
LAN-8	2 hosts

and base address, 175.170.32.0/19

To facilitate 3000 hosts we need 12 host bits as $2^{12} - 2 = 4096 - 2 = 4094$

∴ possible subnets:—

- 1) 175.170.32.0/20 (assigned to LAN-4)
- 2) 175.170.48.0/20

Now base address will be considered as 175.170.48.0/20.

for 1500 hosts we need 11 host bits

∴ possible subnets:—

- 1) 175.170.48.0/21 (assigned to LAN-6)
- 2) 175.170.56.0/21

Now new base address 175.170.56.0/21

Number of hosts 1000

bits required 10 for hosts

∴ possible subnets:-

1) 175.170.56.0/22 (Assigned to LAN-3)

2) 175.170.60.0/22

New base address considered 175.170.60.0/22

Number of hosts 80

bits required 7 for hosts

∴ possible subnets:-

1. 175.170.60.0/25 (Assigned to LAN-2)

2. 175.170.60.128/25

3. 175.170.61.0/25

∴

Now, taking 175.170.60.128/25 as base address

number of hosts 20.

number of host bits required = 5

∴ possible subnets:-

1) 175.170.60.128/27 assigned to LAN-5

2) 175.170.60.160/27

3) 175.170.60.192/27

4) 175.170.60.224/27

Now, remaining network address 175.170.60.160/2

number of hosts 10.

number of bits required for host is 4.

∴ possible subnets:-

1) 175.170.60.176/28 (for LAN-1)

2) 175.170.60.192/28

Now for LAN-7 and LAN-8 we need 2 subnets each having 2 hosts.

∴ number of host bits required 2.

∴ possible subnets:-

1) 175.170.60.196/30 (LAN-7)

2) 175.170.60.200/30 (LAN-8)

3) 175.170.60.204/30

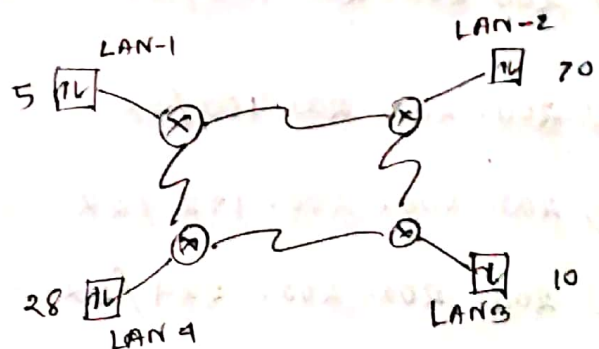
4) 175.170.60.208/30

math

Subnet

1

1. LAN 1: 5 hosts
- LAN 2: 70 hosts
- LAN 3: 10 hosts
- LAN 4: 28 hosts



and 4 serial network of 2 hosts each.

Given base address 200.200.200.0/24

Here, maximum number of hosts is 70

To facilitate ~~number of~~ 70 hosts, we need
7 bits for hosts as $2^7 - 2 = 126 > 70$

possible subnets with 7 bits for hosts -

1. 200.200.200.0000 0000 /25

2. 200.200.200.1000 0000 /25

here ip 200.200.200.0/25 is assigned to LAN 2

For LAN-4, having 28 hosts,

we need $(2^5 - 2 > 28)$, so 5 bits for host position.

• possible subnets are -

① 200.200.200.128/27 - assigned to LAN-4

② 200.200.200.160/27

$\begin{array}{r} 10100000 \\ 11100000 \end{array}$

③ 200.200.200.192/27

④ 200.200.200.224/27

Now, 200.200.200.160/27 will be used to create subnets

For LAN-3, we have 10 hosts.

∴ 4 host bit is required ($2^4 - 2 > 10$)

∴ possible subnets:

① 200.200.200.160/28 - assigned to LAN-3

② 200.200.200.176/28

Now, 200.200.200.176/28 will be used for subnetting.

For LAN-1, 5 hosts, so 3 bits are required ($2^3 - 2 > 5$)

∴ possible subnets:

① 200.200.200.176/29 - assigned to LAN-1

② 200.200.200.184/29

Now, request host per subnet 2

so, 2 bits for 2 hosts

① 200.200.200.184/30 — (Serial-1)

② 200.200.200.188/30 — (Serial-2)

Now from 200.200.200.192/27

① 200.200.200.192/30 — (Serial-3)

② 200.200.200.196/30 — (Serial-4)

(An)

Q: LAN-1: 10 hosts

LAN-2: 80 "

LAN-3: 1000 "

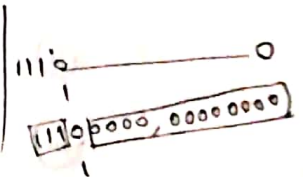
LAN-4: 3000 "

LAN 5: ²⁰1500 hosts

LAN 6: 1500 "

LAN 7: 2 hosts

LAN 8: 2 "



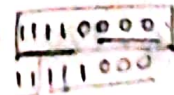
And base address, 175.170.32.0/19

Ans: To facilitate 3000 hosts, we need 12 host bit

∴ possible subnet:

① 175.170.32.0/20 — assigned to LAN-4

② 175.170.48.0/20



For 1500 hosts we need 11 bits

∴ possible subnets -

① 175.170.48.0/21 (assigned LAN-5)

② 175.170.56.0/21

Now base address : 175.170.56.0/21

for 1000 hosts, bits 10

∴ possible subnets —

① 175.170.56.0/22 (assigned to LAN 3)

② 175.170.60.0/22

Now, number of host 80, required bits 7

∴ possible subnet:

① 175.170.60.0/25 — (assign LAN-2)

② 175.170.60.128/25

③ 175.170.61.0/25

⋮

Now, number of hosts 20, required bits 5

∴ possible subnets —

① 175.170.60.128/27 = assign to LAN-5

② 175.170.60.160/27

③ 175.170.60.192/27

④ 175.170.60.224/27

Number of host 10, $\therefore$ bits 4

$\therefore$ possible subnet -

① 175. 170. 60. 176 /28 (for LAN-1)

② 175. 170. 60. 192 /28

for LAN-7 & LAN-8, we need 2 subnets
each having 2 hosts. $\therefore$ bit required 2

$\therefore$ possible subnet -

① 175. 170. 60. 196 /30 - (LAN-7)

② 175. 170. 60. 200 /30 - (LAN-8)

③ 175. 170. 60. 204 /30

④ 175. 170. 60. 208 /30

Given IP address : 172.16.0.0/16

To create 100 subnets, we require 7 bits extra for network

∴ 1st subnet : 172.16.0.0/23

2. total subnet = $2^7 = 128$

2. New subnet mask = 255.255.254.0

3. Hosts per subnet = $2^{32-23} - 2$
 $= 2^9 - 2$
 $= 512 - 2 = 510$

4. total hosts = $128 \times 510 = 65280$

Number of hosts if not been subnetting,

$$2^{32-16} - 2 = 2^{16} - 2 = 65534$$

∴ host losses = $65534 - 65280$

$$= 254 \quad (Ans)$$

IP address: 20.69.0.0/10

subnet the network so that each subnet has at least 3600 hosts.

① How many subnets?

② Consider subnet no. 373, find

(a) Network address of the subnet

(b) First, Last host address

(c) Broadcast address

(d) Unnecessary host addresses in the subnet

③ How many host address losses for subnetting?

Sol

① for 3600 hosts, require $2^{12} = 2 > 3600$

$\therefore$ Bits remaining for subnetting = $32 - 10 - 12 = 10$

$\therefore$ Total subnet = $2^{10} = 1024$

② Binary for 373 is 101110101 (9 bits)

(a)

But 10 places for subnet bits

Putting 0 before binary we have

20.01010111,01010000.00000000

or 20.87.80.0/20 $\rightarrow$ network address

(b) First host address 20.87.80.1/20

Broadcast address 20.01010111.01011111.11111111

or 20.87.95.255/20

$\therefore$ Last host address 20.87.95.254/20

(c) Broadcast address: 20.87.95.255/20

(d) Host ip in each subnet = $2^{12} - 2$

$= 4094$

$\therefore$ Unnecessary host ip = $4094 - 3600 = 494$

(3) Total host ip in original net = $2^{22} - 2 = 4194302$

Host ip in subnet = $2^{10} \times 2^{12} \cdot 4094$

$= 4192256$

$\therefore$ Losses = $4194302 - 4192256$

$= 2046$

(B)

Determine the header checksum of IPv4 packet

i) HLEN = 6 ii) TOS = 2 iii) Total length = 30, iv) identity = 15,

v) Flag = 0, vi) offset = 0, vii) TTL = 5 viii) Protocol = UDP → 17
TCP → 6

ix) Source IP: 10.10.109.5 x) Destination IP: 192.168.2.100

Soln:

4	6	2	30
15	0	0	
5	17	0	
10.10.109.5			
192.168.2.100			

4, 6 & 2 → 0100 0110 000000010

30 → 0000 0000 00011110

15 → 0000 0000 00001111

0 & 0 → 0000 0000 00000000

5 & 17 → 0000 0101 00010001

0 → 00 00 0000 00000000

10.10 → 0000 1010 00001010

109.5 → 0110 1101 00000101

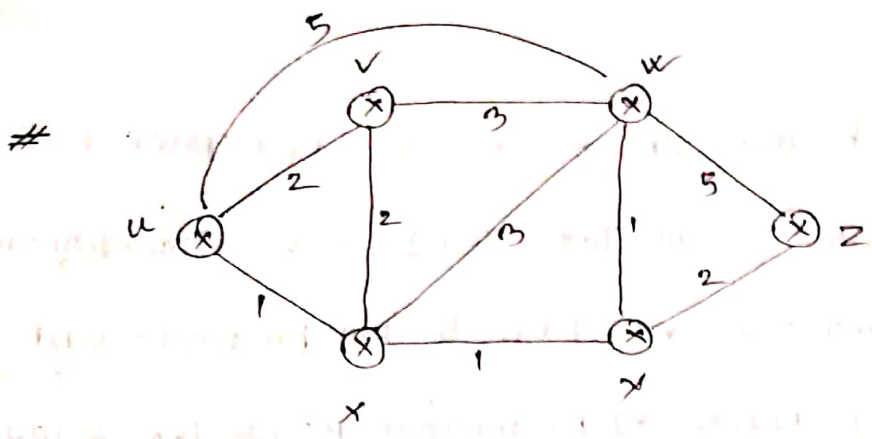
192.168 → 1100 0000 10101000

2.100 → 0000 0010 01100100

Sum → 011 000 1001 011011

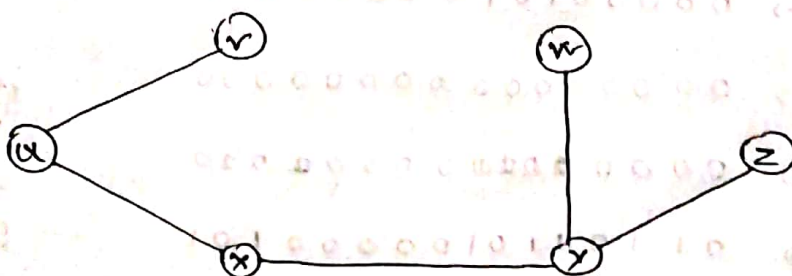
checksum 100 1110 1101 00100

+1
11
11
000
00



Step	N'	$D(v), P(v)$	$D(w), P(w)$	$D(x), P(x)$	$D(y), P(y)$	$D(z), P(z)$
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x		2, x	∞
2	uxy	2, u	3, y			4, y
3	uxyv		3, y			4, y
4	uxyvw					4, y
5	uxyvwz					

Resulting shortest path tree from u -



Resulting forwarding table:

destination	link
v	(u, v)
x	(u, x)
y	(u, x)
w	(u, x)
z	(u, x)

Q: Using RSA algorithm, encrypt the message 'F' and decrypt the cipher text to retrieve the original message.

[choose $p=5$, $q=7$ to find out public & private key]

Solⁿ: Given, $p=5$, $q=7$

$$\text{we know, } n = pq = 5 \times 7 = 35$$

$$\phi = (p-1)(q-1) = 4 \times 6 = 24$$

Let, $e=5$, since 5 & 24 have no common factors

and $d=29$ since $(5 \cdot 29 - 1)$ or 144 is exactly divisible by 24.

$\therefore$ public key $(m, e) = (35, 5)$, private key $(m, d) = (35, 29)$

For encryption:

Letter	m	m^e	Ciphertext $c = m^e \bmod n$
F	6	7776	6

For decryption

ciphertext c	c^d	$m = c^d \bmod n$	Letter
6	$3.084565329 \times 10^{22}$	6	F

(47)

Masud Sir math (A)

~~Host A~~ How long it takes to send a file of 640,000 bits from Host A to B. over a circuit switched network.

.. Network use TDM with 24 slots and having a bit rate of 1.536 mbps.

It takes .500 msec to establish an end to end circuit ; How long does it take to send the file?

Soln;

Each circuit has a transmission rate of

$$\frac{1.536 \text{ mbps}}{24} = .64 \text{ kbps}$$

So, it takes $\frac{640,000}{.64}$ or 10 seconds to transmit the file.

$$\begin{aligned} \text{Adding circuit establishment time} &= (10 + 0.5) \text{ sec} \\ &= 10.5 \text{ sec} \end{aligned}$$

to send a file.

(R)

Packet switching

• send a packet of L bits from A to B

• 2 Links between 2 hosts (A, B), each of rate R bps.

• to transmit first packet, time = $\frac{L}{R}$

$$\text{total Delay} = \frac{2L}{R}$$

2. a

object size = 2×10^5 bits

$\therefore$ size = 0.2 mbits

$$\therefore \text{LAN delay} = \frac{7 \times 0.2}{1000} = 1.4 \times 10^{-3} \text{ sec}$$

$$\text{Access delay} = \frac{7 \times 0.2}{1.54} = 0.91 \text{ sec}$$

Internet delay = 3 sec

$$\begin{aligned} \therefore \text{total average response time} &= (1.4 \times 10^{-3} + 0.91 + 3) \text{ sec} \\ &= 3.9114 \text{ sec} \end{aligned}$$

- c) Suppose, your web browser requests for a webpage which contains 7 small objects in the same server (8 objects including the base HTML file). RTT_0 denotes the RTT between two end systems. Also, pipelining is not considered. How much time is required in terms of RTT_0 from when the client clicks on the link until the client receives the full webpage considering only propagation delay (RTT_0)? (12)
- i) Using non-persistent HTTP, ii) Using persistent HTTP.

2. @

file t_u will be 0

~~8 RTT = Non-persistent~~

Persistent:

2 RTT + file transmission

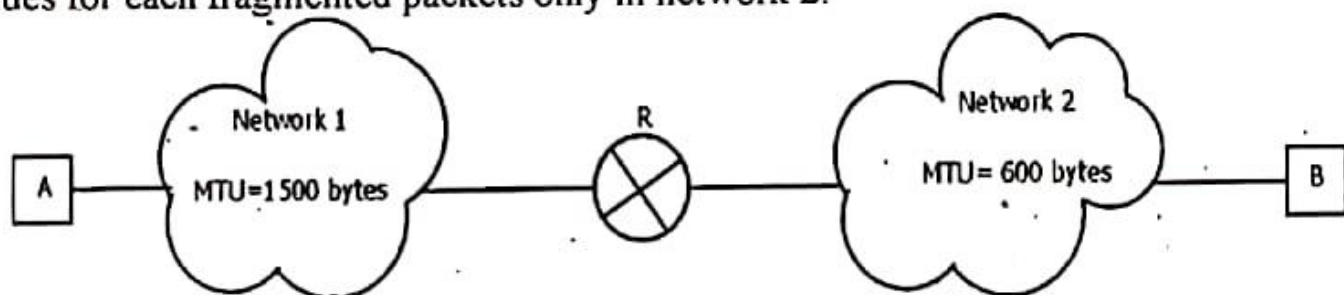
= ~~2~~ 2x8 + 0 $\rightarrow$ no (bottleneck
time
transit)

Non-persistent

RTT + file t_u

= 8 + 1

- d) For the figure in Fig. 1(d), station A needs to send a payload size of 1400 bytes (including 20 bytes of header) to station B across the network as shown. Data needs to be fragmented as the payload is too big to fit in the smallest MTU size (600 bytes). Find the total data length, identification number, more flag and fragmentation offset values for each fragmented packets only in network 2.



- J J J J 1 0 60 120 1 1 1 ^{P20}
 180 240 300 360

[20 | 1380]

[20 | 580]

$$P_1 = 20 + 580 \quad P_2 = 20 + 580 \quad P_3 = 20 + 160$$

TL	P_1	P_2	P_3
MF	20+580	20+580	20+160
offset	1	1	0
	0	72.5	145

class-full

An IP address allocation method that allocates IP addresses into 5 major classes.

Less practical and wasteful

Network and host id changes according to class

consume more bw. / use hello msg ^{don't}

add 3 parts $\rightarrow$ network, subnet, host

class-less

An IP address allocation method that is designed to replace classful addressing to minimize rapid exhaustion of IP add.

more practical and useful

There is no boundary on network and host id.

consume less bw / use hello msg

2 parts $\rightarrow$ subnet / prefix
host